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Safety Control with Adaptive Safety Sets and Belief Dynamics for LAV Under Uncertainty

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Highlights

What are the main findings?

- A belief-driven safety control algorithm is proposed that employs Variational Inference to project non-Gaussian measurement likelihoods into a tractable Gaussian belief. This enables the construction of adaptive buffered half-space constraints, transforming complex probabilistic collision avoidance into deterministic convex geometric checks with rigorous bounds on collision probability.
- An event-triggered hierarchical planning architecture is developed to balance global optimality and real-time responsiveness. By activating global trajectory optimization only upon detecting new obstacles or reaching local goals, the framework avoids the computational burden of continuous replanning while effectively escaping local optima in dense, unknown environments.

What are the implications of the main findings?

- The proposed framework effectively resolves the trade-off between probabilistic safety under non-Gaussian uncertainties and computational efficiency, offering a scalable solution for Loitering Aerial Vehicle (LAV) missions in complex scenarios with mixed known/unknown obstacles.
- This technical paradigm lays a solid foundation for the practical deployment of LAVs in aerial surveying, emergency rescue, and communication relay missions, thereby enhancing operational robustness in dynamic and uncertain environments.

Abstract

This paper presents a novel safety control framework for Loitering Aerial Vehicles (LAVs) operating under non-Gaussian measurement uncertainty. The approach integrates variational inference-based belief dynamics with adaptive buffered half-space constraints, transforming complex probabilistic collision avoidance into tractable convex geometric conditions. This ensures rigorous safety guarantees while avoiding the conservatism of robust methods. An event-triggered hierarchical planner further balances global optimality with local responsiveness, enabling rapid navigation in dynamic environments. Validated through 1000 Monte Carlo simulations, the framework achieves a 95.4% success rate. Comparative analysis demonstrates that the proposed method compares favorably with state-of-the-art safety-set approaches by effectively resolving local infeasibility issues and maintaining real-time efficiency without compromising probabilistic safety assurance.

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Academic Editor: Pablo Rodríguez-González

Received: 31 January 2026
Revised: 1 April 2026
Accepted: 2 April 2026
Published: 3 April 2026

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Keywords: loitering aerial vehicles; obstacle avoidance; non-Gaussian uncertainty; adaptive safety set; belief dynamics

1. Introduction

Loitering Aerial Vehicles (LAVs), a distinctive class of unmanned aerial systems endowed with advanced sensing and long-endurance capabilities, have garnered widespread attention across diverse applications. These include aerial surveying, emergency rescue, security patrol, communication relay, and material delivery, driven by their cost-effectiveness, flexible deployment, and extended loitering performance [1–4]. However, in practical missions, LAVs often operate in complex environments populated by four types of obstacles: known and unknown, both stationary and moving. Constrained by onboard sensor measurements (e.g., relative distance and bearing) for state estimation, LAVs inevitably encounter non-Gaussian uncertainties arising from measurement noise. Furthermore, the reliance on onboard sensors for the dynamic discovery of unknown obstacles necessitates safety control algorithms capable of rapid response to emerging hazards. When coupled with limited sensing ranges and the stochastic nature of moving obstacles, these challenges severely impede reliable obstacle avoidance and safe operations. Consequently, developing a novel safety control algorithm tailored to such complex and uncertain scenarios is an urgent necessity, forming the core motivation of this research.

Existing obstacle avoidance strategies for autonomous vehicles can be broadly categorized into three paradigms: reactive collision avoidance [5–8], planning-based approaches [9–13], and Model Predictive Control (MPC) methods [14–16]. Planning-based methods leverage prior environmental knowledge and real-time situational awareness to generate collision-free trajectories, whereas reactive methods directly map environmental perceptions to avoidance maneuvers. In contrast, MPC-based approaches explicitly incorporate collision avoidance constraints into an optimization framework to derive feasible control inputs. Despite their individual strengths, these methods often face challenges in balancing reliability, computational efficiency, and adaptability when simultaneously addressing non-Gaussian uncertainties, limited sensing ranges, and dynamic unknown obstacles.

The reliable execution of obstacle avoidance maneuvers critically hinges on the accuracy of estimated obstacle states, such as position and velocity. Bayesian filtering offers a unified probabilistic framework for this task, recursively updating the posterior state distribution by fusing prior system knowledge with observational data via Bayes' theorem [17–20]. Under assumptions of linear dynamics and Gaussian noise, the Kalman Filter (KF) provides an optimal closed-form solution. For nonlinear systems, variants like the Extended Kalman Filter (EKF), Unscented Kalman Filter (UKF), and Particle Filter (PF) are widely employed, utilizing distinct approximation strategies to address nonlinearities. Nevertheless, significant challenges remain. The presence of non-Gaussian measurement noise in LAV applications violates the fundamental Gaussian assumptions of KF, EKF, and UKF, leading to inaccurate uncertainty quantification. Conversely, while PF can theoretically handle non-Gaussian distributions, they often incur prohibitive computational costs, rendering them unsuitable for resource-constrained LAVs. Consequently, these limitations make traditional state estimation methods inadequate for ensuring reliable safety control in complex operational scenarios.

To mitigate challenges induced by uncertainties, various robust and stochastic control strategies have been developed. Among them, Tube MPC [21] has gained prominence by decoupling the system dynamics into a nominal trajectory and an error bound. This

https://doi.org/10.3390/drones10040260

approach tightens constraints using robust invariant sets and employs an ancillary feedback controller to confine the uncertain system states within a predefined 'tube'. Notable extensions include tube-based robust MPC for multi-dimensional uncertainties [22], hierarchical virtual tube methods for swarm navigation [23], and MPC frameworks integrating safety sets with tube constraints for aerial obstacle avoidance [24]. However, a significant limitation persists: Tube MPC typically necessitates a sufficiently long prediction horizon to ensure recursive stability. This requirement drastically increases the computational burden, thereby hindering real-time implementation—a critical constraint for LAVs operating in fast-changing dynamic environments.

In contrast to robust methods that rely on conservative worst-case assumptions, Stochastic MPC exploits known uncertainty distributions to permit constraint violations within prespecified probability thresholds. This approach effectively balances safety and performance by reducing unnecessary conservatism. Notable contributions include chance-constrained MPC for autonomous navigation [25], convex chance-constraint-based safety filters [26], and buffered uncertainty-aware Voronoi cells for probabilistic collision avoidance [27]. However, a fundamental bottleneck persists: the handling of probabilistic constraints. While efficient analytical transformations exist for Gaussian uncertainties, deriving deterministic equivalents under non-Gaussian distributions is often computationally intractable. This limitation severely restricts the applicability of current Stochastic MPC frameworks to LAV scenarios characterized by complex, non-Gaussian measurement noise.

Distinct from MPC-based approaches, Belief Space Planning (BSP) operates directly in the space of belief states—defined as posterior probability distributions over system states—thereby explicitly incorporating state estimation uncertainty into the decision-making process. Formulated typically as a Partially Observable Markov Decision Process (POMDP) [28], BSP updates belief dynamics via Bayesian filtering to maintain full posterior representations. Since exact POMDP solutions are computationally intractable, local trajectory optimization algorithms, such as Iterative Linear Quadratic Gaussian (iLQG) [17] and its constrained variant (CILQG) [11], are widely employed. In obstacle-rich environments, these planners enforce safety by evaluating probabilistic collision constraints, often approximating the interaction between belief distributions and obstacle regions (e.g., via potential functions [29]). Nevertheless, significant challenges remain: existing BSP methods frequently incur prohibitive computational costs or struggle to accommodate non-Gaussian uncertainties. These limitations hinder their practical deployment on resource-constrained LAVs operating in complex, dynamic environments.

Motivated by the aforementioned limitations, this paper proposes a novel safety control framework for LAVs operating under non-Gaussian measurement uncertainty. In contrast to traditional Stochastic MPC or BSP approaches—which often suffer from computationally intractable probabilistic integrals or overly conservative Gaussian approximations—the proposed method fundamentally redefines the safety verification problem. By leveraging belief dynamics to construct explicit, adaptive convex safety sets, the framework transforms the problem from enforcing complex, non-convex chance constraints into ensuring the geometric separation of convex sets. This transformation enables high-probability safety guarantees even in the presence of non-Gaussian noise. The main contributions of this work are summarized as follows:

1. A belief-driven safety formulation for non-Gaussian uncertainties: Unlike Robust Tube MPC [24], which relies on conservative worst-case bounding, or Buffered Voronoi methods [27], which depend on restrictive Gaussian assumptions, this work proposes a systematic approach to construct adaptive buffered half-space constraints. By employing Variational Inference (VI) to project non-Gaussian measurement likelihoods into a tractable, single-Gaussian belief, the proposed method reformulates the inher-

ently difficult probabilistic collision avoidance condition as a deterministic convex geometric constraint with statistically derived buffers. This theoretical reformulation enables explicit safety guarantees under non-Gaussian noise, eliminating both the excessive conservatism of robust tubes and the computational intractability of general stochastic integration.

2. An event-triggered hierarchical architecture for scalable planning: A dual-layer framework is designed to dynamically balance near-global optimality with local responsiveness. To address the myopic behavior of local planners that often traps systems in local optima amidst unknown obstacles, an event-triggered mechanism activates global trajectory optimization only when necessary (e.g., upon detecting new obstacles or reaching local goals). This strategy enables LAVs to rapidly navigate around both known and unknown obstacles while preserving real-time performance, offering a scalable solution for dense, dynamic environments without the burden of continuous global replanning.
3. A control-aware belief dynamics modeling pipeline: A VI-based estimation scheme is developed to bridge the gap between complex non-Gaussian sensing and deterministic convex optimization. Unlike standard filters that may yield multi-modal or intractable posterior distributions, the presented method explicitly projects non-Gaussian measurement likelihoods into a single unimodal Gaussian belief. Comparative analysis with Multi-Model EKF demonstrates that this approximation retains comparable estimation fidelity while providing a control-compatible representation. This structural alignment ensures that the resulting belief can be directly embedded into adaptive half-space constraints, enabling computationally efficient probabilistic safety enforcement without resorting to expensive sampling or overly conservative bounding techniques.

The remainder of this paper is organized as follows: Section 2 formulates the safety control problem for LAVs. Section 3 details the proposed safety control algorithm integrating adaptive safety sets and belief dynamics. Section 4 verifies the algorithm's effectiveness via simulation experiments. Section 5 concludes the work and discusses future research directions.

2. Statement of Safety Control Problems

2.1. LAV Dynamic Model

The dynamics of the LAV system in the loitering phase are modeled by

$$\begin{cases} \dot{x}(t) = V(t)\cos\gamma(t)\cos\psi(t) \\ \dot{y}(t) = V(t)\cos\gamma(t)\sin\psi(t) \\ \dot{h}(t) = V(t)\sin\gamma(t) \\ \dot{V}(t) = a_x(t) - g\sin\gamma(t) \\ \dot{\gamma}(t) = \big(a_y(t) - g\cos\gamma(t)\big)/V(t) \\ \dot{\psi}(t) = a_h(t)/\big(V(t)\cos\gamma(t)\big) \end{cases} \tag{1}$$

where $\left[x(t), y(t), h(t)\right]^\top$ represents the position of the LAV in the East-North-Up (ENU) coordinate frame, and $V(t)$ is the airspeed. The angle between the velocity vector and the x-y plane is denoted by $\gamma(t)$, while $\psi(t)$ indicates the angle between the projection of the velocity vector onto the x-y plane and the east direction, as shown in Figure 1. The acceleration of the LAV is given by $\left[a_x(t), a_y(t), a_h(t)\right]^\top$, and g is the gravitational acceleration. Here, t denotes time.

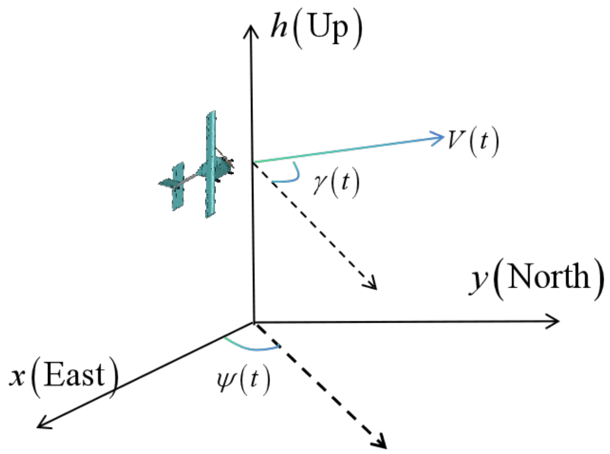

Figure 1. Definition of angles $\gamma(t)$ and $\psi(t)$ in the ENU coordinate system.

A linear prediction model over a short horizon is given as follows

$$\dot{\boldsymbol{X}}(t) = \boldsymbol{A}\boldsymbol{X}(t) + \boldsymbol{B}\boldsymbol{u}(t) \tag{2}$$

where $\boldsymbol{X}(t) = \left[x(t), y(t), h(t), \dot{x}(t), \dot{y}(t), \dot{h}(t)\right]^\top$ and $\boldsymbol{u}(t) = \left[u_x(t), u_y(t), u_h(t)\right]^\top$ denote the state and input vectors of the linear system, respectively. The control input is described by

$$\begin{bmatrix} u_x(t) \\ u_y(t) \\ u_h(t) \end{bmatrix} = \begin{bmatrix} \cos\gamma(t)\cos\psi(t) & -\sin\gamma(t)\cos\psi(t) & -\cos\gamma(t)\sin\psi(t) \\ \cos\gamma(t)\sin\psi(t) & -\sin\gamma(t)\sin\psi(t) & \cos\gamma(t)\cos\psi(t) \\ \sin\gamma(t) & \cos\gamma(t) & 0 \end{bmatrix} \begin{bmatrix} a_x(t) - g\sin\gamma(t) \\ a_y(t) - g\cos\gamma(t) \\ a_h(t)/\cos\gamma(t) \end{bmatrix}. \tag{3}$$

The system matrices are described by

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{0}_{3\times3} & \boldsymbol{I}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \end{bmatrix} \in \mathbb{R}^{6\times6}, \quad \boldsymbol{B} = \begin{bmatrix} \boldsymbol{0}_{3\times3} \\ \boldsymbol{I}_{3\times3} \end{bmatrix} \in \mathbb{R}^{6\times3} \tag{4}$$

where $\boldsymbol{I}_{3\times3}$ and $\boldsymbol{0}_{3\times3}$ denote the 3×3 identity and zero matrices, respectively.

Remark 1. *The linear model* (2) *is employed for prediction to ensure convexity and real-time solvability, while the actual state is propagated via the nonlinear dynamics* (1) *to correct prediction biases. This dual-model structure leverages the inherent robustness of the receding horizon mechanism: the mismatch between the linear prediction and nonlinear reality acts as a bounded additive disturbance, which is naturally constrained by the short prediction horizon [24,30]. Theoretical foundations in robust MPC establish that such systems retain Input-to-State Stability provided the disturbance remains bounded and the state feedback is updated frequently [30,31].*

2.2. Probabilistic Obstacle Avoidance Formulation

To capture the uncertainties in obstacle motion, the kinematic model for ground obstacles is formulated using a linear stochastic differential equation

$$\dot{\boldsymbol{X}}_{\mathrm{o}}(t) = \boldsymbol{A}\boldsymbol{X}_{\mathrm{o}}(t) + \boldsymbol{B}\big(\boldsymbol{u}_{\mathrm{o}}(t) + \boldsymbol{w}_{\mathrm{o}}(t)\big) \tag{5}$$

where $\boldsymbol{X}_{\mathrm{o}}(t) = \left[x_{\mathrm{o}}(t), y_{\mathrm{o}}(t), 0, \dot{x}_{\mathrm{o}}(t), \dot{y}_{\mathrm{o}}(t), 0\right]^\top$ represents the obstacle's state vector, $\boldsymbol{u}_{\mathrm{o}}(t) \in \mathbb{R}^{3\times1}$ denotes the input vector, and $\boldsymbol{w}_{\mathrm{o}}(t) \sim \mathcal{N}\big(\boldsymbol{0}_{3\times1}, \boldsymbol{Q}_{\mathrm{o}}(t)\big)$ is Gaussian-distributed process noise with covariance matrix $\boldsymbol{Q}_{\mathrm{o}}(t) \in \mathbb{R}^{3\times3}$. For stationary obstacles, the input vanishes: $\boldsymbol{u}_{\mathrm{o}}(t) \equiv \boldsymbol{0}_{3\times1}$. The subscript "o" refers to "obstacle".

To address uncertainties in obstacle position estimation while avoiding overly conservative maneuvers, the obstacle avoidance constraint is formulated as a chance constraint

$$\mathbb{P}\Big\{f\Big(x(t), y(t), \hat{O}_{\mathrm{o}}^{j}(t)\Big) \geq 0\Big\} \geq 1-\delta_{\mathrm{o}}, \quad j=1,2,\ldots,n \tag{6}$$

where $n = n_{\mathrm{kn}} + n_{\mathrm{un}}$, with n_{kn} and n_{un} denoting the numbers of known and unknown obstacles, respectively; δ_{o} represents the collision probability threshold. The function $f(\cdot)$ defines the obstacle avoidance constraint. Here, $\mathbb{P}$ denotes the probability measure derived from the posterior distribution of obstacle position estimates, and thus $\mathbb{P}\{\cdot\}$ represents the probability that the event within the braces occurs. Specifically, $\hat{O}_{\mathrm{o}}^{j}(t)$ is the projected area of the j-th obstacle on the x-y plane, centered at the estimated obstacle position, with its size characterized by the parameter $\varepsilon_{\mathrm{o}}^{j}$. For circular projections, $\varepsilon_{\mathrm{o}}^{j}$ is defined as the radius R_{o}^{j}. Consequently, $f(\cdot)$ is defined as the distance between the x-y plane projection of the LAV, $\left[x(t), y(t)\right]^{\top}$, and the estimated obstacle position, $\left[\hat{x}_{\mathrm{o}}^{j}(t), \hat{y}_{\mathrm{o}}^{j}(t)\right]^{\top}$, minus the radius R_{o}^{j}

$$f := \left\| \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} - \begin{bmatrix} \hat{x}_{\mathrm{o}}^{j}(t) \\ \hat{y}_{\mathrm{o}}^{j}(t) \end{bmatrix} \right\| - R_{\mathrm{o}}^{j}. \tag{7}$$

For rectangular projections, $\varepsilon_{\mathrm{o}}^{j}$ is specified as the length (in the x-direction) and width (in the y-direction) of the projection. The function $f(\cdot)$ is defined to ensure that the x-y plane projection of the LAV lies outside the rectangular box

$$f := \boldsymbol{A}_{\mathrm{o}}^{j,i} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} - b_{\mathrm{o}}^{j,i}, \quad i=0,1,\ldots,n_{\mathrm{ver}}^{j}-1 \tag{8}$$

where the superscript "i" denotes the i-th vertex of the box (arranged in clockwise order), n_{ver}^{j} is the total number of vertices for the j-th obstacle's bounding box, and the coefficients are derived from the equations of the straight lines forming the box sides

$$\boldsymbol{A}_{\mathrm{o}}^{j,i} = \begin{bmatrix} y_{\mathrm{o}}^{j,i} - y_{\mathrm{o}}^{j,i+1} & x_{\mathrm{o}}^{j,i+1} - x_{\mathrm{o}}^{j,i} \end{bmatrix}, \quad b_{\mathrm{o}}^{j,i} = x_{\mathrm{o}}^{j,i+1} y_{\mathrm{o}}^{j,i} - x_{\mathrm{o}}^{j,i} y_{\mathrm{o}}^{j,i+1} \tag{9}$$

where $\left[x_{\mathrm{o}}^{j,i}, y_{\mathrm{o}}^{j,i}\right]^{\top}$ is the position of the i-th vertex of the j-th obstacle.

2.3. Detection Area Modeling for Onboard Sensors

The detection area of onboard sensors is defined as the region where measurements—specifically, the distance and bearing of obstacles relative to the LAV—can be acquired. This area correlates with the LAV's flight altitude and the sensors' field-of-view parameters. In this study, assuming the sensors are installed directly beneath the LAV, the detection area is modeled as

$$F(t) := \left\{ \begin{bmatrix} x_D(t) \\ y_D(t) \end{bmatrix} \,\middle|\, \frac{(x_D(t)-x(t))^2}{L_1^2} + \frac{(y_D(t)-y(t))^2}{L_2^2} \leq 1 \right\} \tag{10}$$

where $L_1 = h(t) \cdot \tan(\alpha)$, $L_2 = h(t) \cdot \tan(\beta)$, $h(t)$ denotes the flight altitude of the LAV, α and β represent the frontal and lateral fields of view of the sensors, respectively, and $\left[x_D(t), y_D(t)\right]^{\top}$ denotes a point within the detection area (including its boundary), as illustrated in Figure 2.

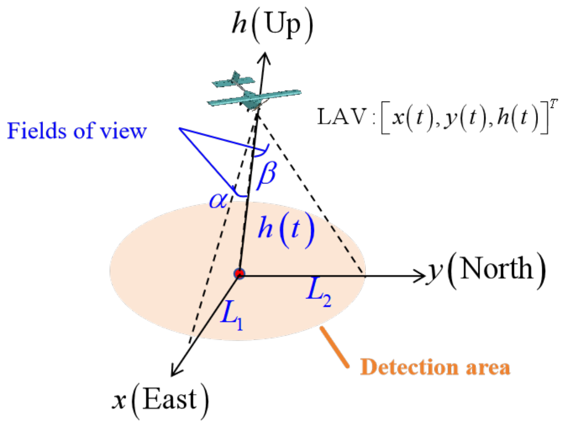

Figure 2. Schematic projection of the onboard sensor detection area on the x-y plane.

2.4. Formulation of Safety Control Problem

The safety control of LAVs is formulated as the following optimization problem

$$\begin{aligned} \min_{\boldsymbol{u}(t)} \ & \underbrace{\left(\boldsymbol{X}(t_f)-\boldsymbol{X}_{\mathrm{d}}\right)^{\top}\boldsymbol{Q}\left(\boldsymbol{X}(t_f)-\boldsymbol{X}_{\mathrm{d}}\right)}_{\text{Terminal Cost}}+\underbrace{\frac{1}{2}\int_0^{t_f}\boldsymbol{u}^{\top}(t)\boldsymbol{R}\boldsymbol{u}(t)\,\mathrm{d}t}_{\text{Control Effort}} \\ \text{s.t.} \ & \dot{\boldsymbol{X}}(t)=\boldsymbol{A}\boldsymbol{X}(t)+\boldsymbol{B}\boldsymbol{u}(t), \\ & \boldsymbol{X}(0)=\boldsymbol{X}_0, \\ & \boldsymbol{X}(t)\in\chi, \\ & \boldsymbol{u}(t)\in\mathcal{U}, \quad \forall t\in[0,t_f], \\ & \left[x(t),y(t),h(t)\right]^{\top}\in\chi_p, \\ & \mathbb{P}\left\{f\left(x(t),y(t),\hat{O}_{\mathrm{o}}^{j}(t)\right)\geq 0\right\}\geq 1-\delta_{\mathrm{o}}, \quad \forall j\in\{1,\ldots,n\}. \end{aligned} \tag{11}$$

where $\boldsymbol{Q}\in\mathbb{R}^{6\times 6}$ and $\boldsymbol{R}\in\mathbb{R}^{3\times 3}$ are symmetric positive definite matrices, and $t_f>0$ denotes the flight duration. The set χ_p represents the permissible mission airspace, which is assumed to be a closed convex set, while χ and $\mathcal{U}$ denote the permissible state and input spaces, respectively, both of which are defined as compact convex sets. $\boldsymbol{X}_0$ and $\boldsymbol{X}_{\mathrm{d}}$ are the initial and target states.

The solution to the optimization problem in (11) faces four principal challenges: (i) the non-convex nature of the obstacle avoidance constraints; (ii) the existence of unknown obstacles within the mission airspace; (iii) the limited reactive capability imposed by the finite detection area of the onboard sensors; and (iv) the presence of non-Gaussian uncertainties stemming from measurement noise in obstacle position estimation. These challenges collectively necessitate a novel algorithmic framework.

3. Safety Control Algorithm Against Known and Unknown Obstacles

This section proposes a safety control algorithm that synergizes belief dynamics with adaptive safety sets. The approach operates through three stages: First, online global trajectory optimization generates a reference path using known obstacles and real-time states. Second, upon obstacle detection, a triggering mechanism updates the belief and safety sets to ensure robustness against state uncertainties. Finally, local targets are dynamically refined to harmonize global guidance with immediate avoidance maneuvers.

3.1. Global Trajectory Optimization for Known Obstacles

The global trajectory from the LAV's state at time t to the target state is generated using prior information about known obstacles, as outlined in Algorithm 1. Let $S_V(t)$ denote the set of state nodes and $S_E(t)$ denote their connecting edges; the global trajectory $\mathcal{T}(t)$ is thus constructed from $S_V(t)$ and $S_E(t)$. The node generation strategy (Line 6) ensures that all nodes remain within the mission airspace, while the collision detection strategy (Line 8) guarantees avoidance of known obstacles under deterministic assumptions.

Algorithm 1 Global Trajectory Optimization

1: **Initialization:**
2: $\mathcal{S}_V(t) \leftarrow \emptyset, \mathcal{S}_E(t) \leftarrow \emptyset$;
3: $\mathcal{S}_V(t) \leftarrow \{\boldsymbol{X}(t)\}$;
4: **Procedure:**
5: **for** $s \in [1, S_{\max}]$ **do**
6: $\quad \boldsymbol{X}_s \leftarrow \left\{ \boldsymbol{X}(t) \mid \boldsymbol{X}(t) \in \chi, \boldsymbol{M}_3\boldsymbol{X}(t) \in \chi_p, \boldsymbol{M}_2\boldsymbol{X}(t) \notin \hat{\mathrm{O}}_\mathrm{o}^j(t), j = 1, 2, \ldots, n_{\mathrm{kn}} \right\}$;
7: $\quad \boldsymbol{X}_f \leftarrow \left\{ \boldsymbol{X} \in \mathcal{S}_V(t) \mid \arg\min_{\boldsymbol{X}} c^*[\boldsymbol{X}, \boldsymbol{X}_s] \right\}$;
8: $\quad$ **if** $\left(\boldsymbol{M}_2 \cdot T^*[\boldsymbol{X}_f, \boldsymbol{X}_s]\right) \notin \hat{\mathrm{O}}_\mathrm{o}^j(t), j = 1, 2, \ldots, n_{\mathrm{kn}}$ **then**
9: $\quad\quad \mathcal{S}_V \leftarrow \mathcal{S}_V \cup \{\boldsymbol{X}_s\}; \mathcal{S}_E \leftarrow \mathcal{S}_E \cup \left\{ (\boldsymbol{X}_f, \boldsymbol{X}_s) \right\}$;
10: $\quad\quad$ **if** $\|\boldsymbol{M}_3(\boldsymbol{X}_s - \boldsymbol{X}_d)\|_2 \leq \epsilon$ **then**
11: $\quad\quad\quad$ **break**;
12: $\quad\quad$ **end if**
13: $\quad$ **end if**
14: **end for**
15: **Return** $\mathcal{T}(t)$.

Let $\boldsymbol{X}_s$ denote the s-th sampling state node, and let its parent node $\boldsymbol{X}_f$ be defined as the state node with the minimum trajectory cost to reach $\boldsymbol{X}_s$ among all candidate nodes. The optimal trajectory from $\boldsymbol{X}_f$ to $\boldsymbol{X}_s$ and the corresponding trajectory cost are given in [3] as

$$T^*[\boldsymbol{X}_f, \boldsymbol{X}_s] = e^{\boldsymbol{A}}\boldsymbol{X}_f - \boldsymbol{G}(\tau^*)\boldsymbol{\lambda}(\tau^*) \tag{12}$$

and

$$c^*[\boldsymbol{X}_f, \boldsymbol{X}_s] = \tau^* + \frac{1}{2}\boldsymbol{d}^\top(\tau^*)\boldsymbol{G}^{-1}(\tau^*)\boldsymbol{d}(\tau^*) \tag{13}$$

where τ^* is the optimal terminal time, $\boldsymbol{\lambda}(\tau^*)$ is the Lagrange multiplier, $\boldsymbol{G}(\tau^*)$ is the Gramian matrix associated with the system matrices, and $\boldsymbol{d}(\tau^*)$ is the difference between the system's terminal state under minimum-energy control and that under zero-control input.

Let $S_{\max}$ represent the maximum sampling count for node generation, and let matrices $\boldsymbol{M}_2$ and $\boldsymbol{M}_3$ extract position information from the state vectors, as given by

$$\boldsymbol{M}_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{M}_3 = \begin{bmatrix} \boldsymbol{I}_{3\times3} & \boldsymbol{0}_{3\times3} \end{bmatrix} \in \mathbb{R}^{3\times6}. \tag{14}$$

3.2. Design an Adaptive Safety Set Under Uncertainty

3.2.1. Belief Dynamics for Relative State Estimation

The adaptive safety set is initially defined as the sensors' detection area, as shown in (10). When an obstacle appears within this area, the LAV estimates its position using onboard measurements. The measurement equation is given by

$$\boldsymbol{z}_t = \bar{\boldsymbol{h}}(\bar{\boldsymbol{X}}_t) + \boldsymbol{v}_t \tag{15}$$

where $\bar{\boldsymbol{X}}_t = \boldsymbol{X}_\mathrm{o}^j(t) - \boldsymbol{X}(t)$ denotes the relative state between the LAV and the j-th obstacle at time t, $\boldsymbol{z}_t$ represents the measurement vector at time t, $\bar{\boldsymbol{h}}(\bar{\boldsymbol{X}}_t)$ is a nonlinear mapping (linearized in Appendix A), and $\boldsymbol{v}_t$ is non-Gaussian measurement noise modeled as a Gaussian Mixture Model (GMM). The probability density function of $\boldsymbol{v}_t$ is

$$p(\boldsymbol{v}_t) = \sum_{m=1}^{M} \alpha_{m,t} \cdot \mathcal{N}(\boldsymbol{\mu}_{m,t}, \boldsymbol{R}_{m,t}) \tag{16}$$

where M denotes the number of Gaussian components in the mixture, $\mathcal{N}(\boldsymbol{\mu}_{m,t}, \boldsymbol{R}_{m,t})$ represents the m-th Gaussian distribution, and $\alpha_{m,t}$ is the mixing weight satisfying $\sum_{m=1}^{M} \alpha_{m,t} = 1$. Here, $\boldsymbol{\mu}_{m,t}$ and $\boldsymbol{R}_{m,t}$ are the mean vector and covariance matrix of the m-th Gaussian component at time t, respectively.

The posterior belief of the relative state is propagated through belief dynamics

$$b_t(\bar{\boldsymbol{X}}_t) \propto p(\boldsymbol{z}_t|\bar{\boldsymbol{X}}_t) \cdot b_{t|t-1}(\bar{\boldsymbol{X}}_t) \tag{17}$$

where $b_t(\bar{\boldsymbol{X}}_t)$ denotes the posterior belief of the relative state $\bar{\boldsymbol{X}}_t$ at time t, $p(\boldsymbol{z}_t|\bar{\boldsymbol{X}}_t)$ is the measurement likelihood, and $b_{t|t-1}(\bar{\boldsymbol{X}}_t)$ represents the prior belief at time t.

Throughout this paper, τ denotes the physical duration of a single discrete time step. Assuming the relative-state belief at time step $t-1$ (corresponding to physical time $t-\tau$) follows a Gaussian distribution with mean $\hat{\bar{\boldsymbol{X}}}_{t-1}$ and covariance $\boldsymbol{\Sigma}_{t-1}$, the a prior prediction $b_{t|t-1}(\bar{\boldsymbol{X}}_t)$ is obtained from the linear models (2) and (5) as

$$\begin{aligned} b_{t|t-1}(\bar{\boldsymbol{X}}_t) &= \mathcal{N}\left(\hat{\bar{\boldsymbol{X}}}_t^-, \boldsymbol{\Sigma}_t^-\right) \\ \hat{\bar{\boldsymbol{X}}}_t^- &= \bar{\boldsymbol{A}}\hat{\bar{\boldsymbol{X}}}_{t-1} + \bar{\boldsymbol{B}}(\boldsymbol{u}_{\mathrm{o},t-1} - \boldsymbol{u}_{t-1}) \\ \boldsymbol{\Sigma}_t^- &= \bar{\boldsymbol{A}}\boldsymbol{\Sigma}_{t-1}\bar{\boldsymbol{A}}^\top + \bar{\boldsymbol{B}}\boldsymbol{Q}_{\mathrm{o},t}\bar{\boldsymbol{B}}^\top \end{aligned} \tag{18}$$

where $\bar{\boldsymbol{A}} = \boldsymbol{I}_{6\times6} + \boldsymbol{A}\tau$, $\boldsymbol{I}_{6\times6} \in \mathbb{R}^{6\times6}$ is an identity matrix, and $\bar{\boldsymbol{B}} = \boldsymbol{B}\tau$. The terms $\boldsymbol{u}_{t-1}$, $\boldsymbol{u}_{\mathrm{o},t-1}$, and $\boldsymbol{Q}_{\mathrm{o},t}$ are abbreviations for $\boldsymbol{u}(t-\tau)$, $\boldsymbol{u}_\mathrm{o}(t-\tau)$, and $\boldsymbol{Q}_\mathrm{o}(t)$, respectively. For unknown obstacles, $\boldsymbol{u}_{\mathrm{o},t-1}$ is set to the zero vector $\boldsymbol{0}_{3\times1}$.

The non-Gaussian measurement noise is modeled using a GMM, which yields the following measurement likelihood

$$p(\boldsymbol{z}_t|\bar{\boldsymbol{X}}_t) = \sum_{m=1}^{M} \alpha_{m,t} \cdot \mathcal{N}\big(\boldsymbol{z}_t; \bar{\boldsymbol{h}}(\bar{\boldsymbol{X}}_t) + \boldsymbol{\mu}_{m,t}, \boldsymbol{R}_{m,t}\big). \tag{19}$$

A latent variable $c_t \in \{1, 2, \ldots, M\}$ is introduced to represent the mixture component assignment for the current measurement. Its distribution is defined as

$$q(c_t = m) = \gamma_{m,t}, \quad \sum_{m=1}^{M} \gamma_{m,t} = 1. \tag{20}$$

The logarithm of the measurement likelihood can be obtained by substituting the variational distribution into (19), yielding

$$\begin{aligned} \log p(\boldsymbol{z}_t|\bar{\boldsymbol{X}}_t) &= \log \sum_{m=1}^{M} p(\boldsymbol{z}_t, c_t = m|\bar{\boldsymbol{X}}_t) \\ &= \log \sum_{m=1}^{M} p(c_t = m) \cdot p(\boldsymbol{z}_t|c_t = m, \bar{\boldsymbol{X}}_t) \\ &= \log\left(\mathbb{E}_{q(c_t)}\left[\frac{p(c_t)p(\boldsymbol{z}_t|c_t, \bar{\boldsymbol{X}}_t)}{q(c_t)}\right]\right). \end{aligned} \tag{21}$$

Applying Jensen's inequality to the logarithm of the expectation in (21) yields the Evidence Lower Bound (ELBO), denoted as $L(q)$

$$\log p(\boldsymbol{z}_t|\bar{\boldsymbol{X}}_t) = \log\left(\mathbb{E}_{q(c_t)}\left[\frac{p(\boldsymbol{z}_t, c_t|\bar{\boldsymbol{X}}_t)}{q(c_t)}\right]\right) \geq \mathbb{E}_{q(c_t)}\left[\log\frac{p(\boldsymbol{z}_t, c_t|\bar{\boldsymbol{X}}_t)}{q(c_t)}\right] \triangleq L(q). \tag{22}$$

Substituting the distributions from (19) and (20), the ELBO can be written as

$$\begin{aligned} L(q) &= \sum_{m=1}^{M} \gamma_{m,t} \log\left(\frac{\alpha_{m,t} \cdot \mathcal{N}(\boldsymbol{z}_t; \bar{h}(\bar{\boldsymbol{X}}_t) + \boldsymbol{\mu}_{m,t}, \boldsymbol{R}_{m,t})}{\gamma_{m,t}}\right) \\ &= \sum_{m=1}^{M} \gamma_{m,t} \log(\alpha_{m,t} \cdot \mathcal{N}(\dots)) - \sum_{m=1}^{M} \gamma_{m,t} \log \gamma_{m,t}. \end{aligned} \tag{23}$$

The optimal variational weights $\gamma_{m,t}$ are obtained by maximizing $L(q)$ subject to $\sum \gamma_{m,t} = 1$

$$\max_{\gamma_{m,t}} L(q) \quad \text{s.t.} \quad \sum_{m=1}^{M} \gamma_{m,t} = 1. \tag{24}$$

This is a standard concave optimization problem over a simplex. By introducing a Lagrange multiplier and solving for the stationary point (detailed derivation provided in Appendix B), the closed-form solution is given

$$\gamma_{m,t} = \frac{\alpha_{m,t} \cdot \exp\left(-\frac{1}{2}\|\boldsymbol{z}_t - \bar{h}(\bar{\boldsymbol{X}}_t) - \boldsymbol{\mu}_{m,t}\|^2_{\boldsymbol{R}^{-1}_{m,t}}\right)}{\sum_{m=1}^{M} \alpha_{m,t} \cdot \exp\left(-\frac{1}{2}\|\boldsymbol{z}_t - \bar{h}(\bar{\boldsymbol{X}}_t) - \boldsymbol{\mu}_{m,t}\|^2_{\boldsymbol{R}^{-1}_{m,t}}\right)}. \tag{25}$$

Since the objective function is concave and the feasible solution space is convex, the derived closed-form expression directly yields the global optimum without the need for iterative convergence.

Using the variational weights $\gamma_{m,t}$, the GMM likelihood in (19) is ultimately approximated by a single Gaussian distribution

$$p(\boldsymbol{z}_t|\bar{\boldsymbol{X}}_t) \approx \mathcal{N}(\bar{\boldsymbol{z}}_t; \bar{\boldsymbol{h}}(\bar{\boldsymbol{X}}_t), \bar{\boldsymbol{R}}_t) \tag{26}$$

$$\bar{\boldsymbol{z}}_t = \sum_{m=1}^{M} \gamma_{m,t}(\boldsymbol{z}_t - \boldsymbol{\mu}_{m,t}), \quad \bar{\boldsymbol{R}}_t = \sum_{m=1}^{M} \gamma_{m,t} \boldsymbol{R}_{m,t}. \tag{27}$$

where $\bar{\boldsymbol{z}}_t$ is derived from bias compensation applied to the raw measurement $\boldsymbol{z}_t$, while $\bar{\boldsymbol{R}}_t$ captures the overall measurement uncertainty.

The Bayesian belief update is performed by combining the prior with the approximated likelihood. Specifically, the posterior belief is proportional to the product of the Gaussian prior (18) and the Gaussian-approximated likelihood (26). Normalizing this product yields the updated belief as a single Gaussian distribution

$$\begin{aligned} b_t(\bar{\boldsymbol{X}}_t) &\propto p(\boldsymbol{z}_t|\bar{\boldsymbol{X}}_t) \cdot b_{t|t-1}(\bar{\boldsymbol{X}}_t) \\ &\approx \mathcal{N}(\bar{\boldsymbol{z}}_t; \bar{\boldsymbol{h}}(\bar{\boldsymbol{X}}_t), \bar{\boldsymbol{R}}_t) \cdot \mathcal{N}\left(\hat{\bar{\boldsymbol{X}}}^-_t, \boldsymbol{\Sigma}^-_t\right) \\ &\approx \mathcal{N}\left(\hat{\bar{\boldsymbol{X}}}_t, \boldsymbol{\Sigma}_t\right) \end{aligned} \tag{28}$$

$$\begin{aligned} \hat{\bar{X}}_t &= \hat{\bar{X}}_t^- + K_t\left(\bar{z}_t - \bar{h}(\hat{\bar{X}}_t^-)\right) \\ \Sigma_t &= (I_{6\times 6} - K_t H_t)\Sigma_t^- \\ K_t &= \Sigma_t^- H_t^\top \left(H_t \Sigma_t^- H_t^\top + \bar{R}_t\right)^{-1} \end{aligned} \tag{29}$$

where H_t is the measurement matrix at time t, and $\hat{\bar{X}}_t$ and Σ_t denote the posterior estimate and covariance of the relative state $\bar{X}_t$ at time t, respectively.

Remark 2. *Convergence and Approximation Limits The variational update process defined by* (25)–(27) *offers distinct advantages for real-time deployment, albeit under specific assumptions. Deterministic Convergence: Unlike iterative variational methods that require multiple steps to converge, our approach derives the optimal variational weights $\gamma_{m,t}$ via the closed-form solution in* (25)*. This guarantees instantaneous convergence to the optimal parameters of the approximating distribution within the chosen family, ensuring numerical stability and predictable computation time. Limitations of Gaussian Approximation: It is crucial to acknowledge that approximating the posterior with a single Gaussian, as in* (28)*, inherently assumes the true belief is unimodal. In scenarios where GMM measurement noise induces distinctly separated multimodal posteriors (e.g., ambiguous measurements arising from symmetric obstacles), this unimodal approximation effectively smooths over the distinct modes. Consequently, this smoothing effect may lead to a failure in detecting critical threats, as the probability mass associated with dangerous modes may be diluted or obscured within the single inflated covariance, rather than being explicitly represented.*

3.2.2. Safety Set Update Under Relative State Uncertainty

The uncertainty of the relative state is projected onto the obstacle space to form the obstacle state belief, denoted as $b_t(X_o^j(t)) \approx \mathcal{N}\left(X(t) + \hat{\bar{X}}_t, \Sigma_t\right))$. Consequently, the obstacle's projected area on the x-y plane is defined as $\hat{O}_b^j(t)$ (termed the belief-propagation obstacle area), centered at the estimated obstacle position derived from the belief.

To guarantee collision-free separation between the initial safety set $F(t)$ (given by (10)) and the belief-propagation obstacle area $\hat{O}_b^j(t)$, the following optimization problem is established

$$\min_{\bar{x}_j(t),\bar{y}_j(t)} \left\| \left[\bar{x}_j(t), \bar{y}_j(t)\right]^\top - \left[x(t), y(t)\right]^\top \right\| \quad \text{s.t.} \quad \left[\bar{x}_j(t), \bar{y}_j(t)\right]^\top \in \hat{O}_b^j(t) \tag{30}$$

where $\left[\bar{x}_j(t), \bar{y}_j(t)\right]^\top$ denotes the nearest point from the LAV to the belief-propagation obstacle area $\hat{O}_b^j(t)$. This point is adopted to construct a separating hyperplane between $F(t)$ and $\hat{O}_b^j(t)$, which is defined on the x-y plane as

$$H_j(t) := \left\{ \left[x_H(t), y_H(t)\right]^\top \,\middle|\, \bar{A}_j\left[x_H(t), y_H(t)\right]^\top = \bar{b}_j \right\} \tag{31}$$

where $\bar{A}_j = \left[x(t) - \bar{x}_j(t), y(t) - \bar{y}_j(t)\right]$ is the normal vector, $\left[x_H(t), y_H(t)\right]^\top$ denotes an arbitrary point on the hyperplane, and $\bar{b}_j = \left(x(t) - \bar{x}_j(t)\right)\bar{x}_j(t) + \left(y(t) - \bar{y}_j(t)\right)\bar{y}_j(t)$ is the offset.

The half-space defined by the normal vector $\bar{A}_j$ is designated as the collision avoidance constraint. Consequently, the initial safety set is adaptively updated via intersection with this constraint, yielding the refined safety set $F(t)$, which guarantees separation from the obstacle

$$F(t) := \left\{ \left[x_D(t), y_D(t)\right]^\top \,\middle|\, \left[x_D(t), y_D(t)\right]^\top \in F(t),\ \bar{A}_j\left[x_D(t), y_D(t)\right]^\top \geq \bar{b}_j - b_j^\delta,\ j = 1, \ldots, n_D(t) \right\} \tag{32}$$

where $n_D(t) \leq n$ is the number of obstacles within the initial safety set at time t. The parameter b_j^δ acts as a probabilistic safety buffer determined by the threshold δ_o, enabling a tunable balance between rigorous safety compliance and reduced algorithmic conservatism. Let $d_j = \bar{\boldsymbol{A}}_j \left[x_D(t), y_D(t)\right]^\top - \bar{b}_j$. When $b_j^\delta = 0$, the constraint $d_j \geq 0$ enforces absolute safety under relative state uncertainty. When $b_j^\delta > 0$, the constraint is relaxed to $d_j \geq -b_j^\delta$, allowing points in $F(t)$ to approach the obstacle more closely, with the permissible proximity strictly governed by the magnitude of b_j^δ.

Given the linearity of the hyperplane, the distance random variable D follows a Gaussian distribution with mean d_j and variance $\sigma^2 = \bar{\boldsymbol{A}}_j \boldsymbol{\Sigma}_t \bar{\boldsymbol{A}}_j^\top$. The collision event is defined as $D < 0$, and the corresponding collision probability is given by

$$\mathbb{P}(D < 0) = \mathbb{P}\left(\frac{D - d_j}{\sigma} < -\frac{d_j}{\sigma}\right) = \Phi\left(-\frac{d_j}{\sigma}\right) \tag{33}$$

where $\Phi(\cdot)$ is the cumulative distribution function (CDF) of the standard normal distribution. To ensure the collision probability remains below the preset threshold δ_o while fully utilizing the probability budget, the following equality constraint is enforced at the boundary

$$\Phi\left(-\frac{d_j}{\sigma}\right) = \delta_o. \tag{34}$$

Solving (34) yields $d_j = -\sigma \cdot \Phi^{-1}(\delta_o)$. By aligning this result with the relaxed safety constraint $d_j \geq -b_j^\delta$, the probabilistic safety buffer is defined as

$$b_j^\delta = \sigma \cdot \Phi^{-1}(1 - \delta_o). \tag{35}$$

Remark 3. *To address potential deviations from Gaussianity in the posterior distribution, Cantelli's inequality (a one-sided Chebyshev inequality) is invoked. This principle provides a conservative, distribution-agnostic safety guarantee for any variable with finite variance, yielding a robust theoretical bound $b_j^{Cheb} = \sigma/\sqrt{\delta_o}$. While this bound ensures safety regardless of distribution shape, it is often overly conservative for approximately Gaussian distributions. Consequently, the proposed framework employs the tighter buffer derived from the Gaussian CDF in (35) to enhance feasibility and performance. The difference $\Delta b_j = b_j^{Cheb} - b_j^\delta$ quantifies the optimistic bias introduced by the Gaussian assumption. In Monte Carlo simulations (1000 runs), the average discrepancy was 10.6 m. This result indicates that while the Gaussian assumption introduces an optimistic bias, its magnitude remains bounded and relatively small within the tested scenarios.*

3.3. Strategies for Local Target Generation

The local optimization framework provides superior dynamic adaptability, enabling real-time responses to unknown and mobile obstacles. To mitigate the inherent shortsightedness of purely local approaches, the globally optimized trajectory from Section 3.1 serves as high-level guidance. Specifically, local targets are defined as states along the global trajectory $\mathcal{T}(t)$ whose projections onto the x-y plane intersect the boundary of the safety set $F(t)$.

$$\boldsymbol{X}_{\mathrm{lg}}(t) := \left\{ \boldsymbol{X}_{\mathrm{lg}}(t) \,\middle|\, \begin{array}{l} \boldsymbol{X}_{\mathrm{lg}}(t) \in \mathcal{T}(t), \left[x_{\mathrm{lg}}(t), y_{\mathrm{lg}}(t)\right]^\top = \boldsymbol{M}_2 \boldsymbol{X}_{\mathrm{lg}}(t), \\ \left(x_{\mathrm{lg}}(t) - x(t)\right)^2 / L_1^2 + \left(y_{\mathrm{lg}}(t) - y(t)\right)^2 / L_2^2 = 1 \quad \text{or} \quad \bar{\boldsymbol{A}}_j \left[x_{\mathrm{lg}}(t), y_{\mathrm{lg}}(t)\right]^\top = \bar{b}_j - b_j^\delta, \\ j = 1, \ldots, n_D(t) \end{array} \right\}. \tag{36}$$

Building on the adaptive safety set and these globally informed local targets, the originally intractable optimization problem in (11) is reformulated into a tractable one

$$
\begin{aligned}
\min_{\boldsymbol{u}_{t|0},\cdots,\boldsymbol{u}_{t|N-1}} \quad & \left(\boldsymbol{X}_{t|N} - \boldsymbol{X}_{\mathrm{lg}}(t)\right)^{\top} \boldsymbol{Q} \left(\boldsymbol{X}_{t|N} - \boldsymbol{X}_{\mathrm{lg}}(t)\right) + \sum_{k=0}^{N-1} \boldsymbol{u}_{t|k}^{\top} \boldsymbol{R} \boldsymbol{u}_{t|k} \\
\text{s.t.} \quad & \boldsymbol{X}_{t|0} = \boldsymbol{X}(t) \\
& \boldsymbol{X}_{t|k+1} = \bar{\boldsymbol{A}} \boldsymbol{X}_{t|k} + \bar{\boldsymbol{B}} \boldsymbol{u}_{t|k} \\
& \boldsymbol{X}_{t|k} \in \chi \\
& \boldsymbol{u}_{t|k} \in \mathcal{U} \\
& \boldsymbol{M}_3 \boldsymbol{X}_{t|k} \in \chi_p \\
& \boldsymbol{M}_2 \boldsymbol{X}_{t|k} \in F(t)
\end{aligned} \tag{37}
$$

where the subscript "$t|k$" denotes the k-th step within an N-step prediction horizon (with step size τ), initiated at time t from the LAV's current state $\boldsymbol{X}(t)$. This simplification leverages the uncertainty-handling capabilities of belief dynamics, the proposed hierarchical planning framework, and the convexity of the adaptive safety set (proven in Appendix C).

The complete implementation of the proposed algorithm is summarized in Algorithm 2, with its workflow illustrated in Figure 3.

Algorithm 2 Safety Control with Adaptive Safety Sets and Belief Dynamics for LAV under Uncertainty

1: **If** (LAV reaches the local target) or (LAV detects unknown obstacles) **then**
2: Optimize the global trajectory via Algorithm 1;
3: **End if**
4: Design the initial safety set according to (10);
5: **for** $j \in n$ **do**
6: **if** $\left[\hat{x}_{\mathrm{o}}^{j}(t), \hat{y}_{\mathrm{o}}^{j}(t)\right]^{\top} \in F(t)$ **then**
7: Capture relative state uncertainties via (18), (25)–(26), (28);
8: Construct the separating hyperplane using (30)–(31);
9: Update the adaptive safety set with (32), (35);
10: **end if**
11: **end for**
12: Generate the globally informed local target based on (36);
13: Solve the safety control optimization problem shown in (37) via CasADi's symbolic optimization.

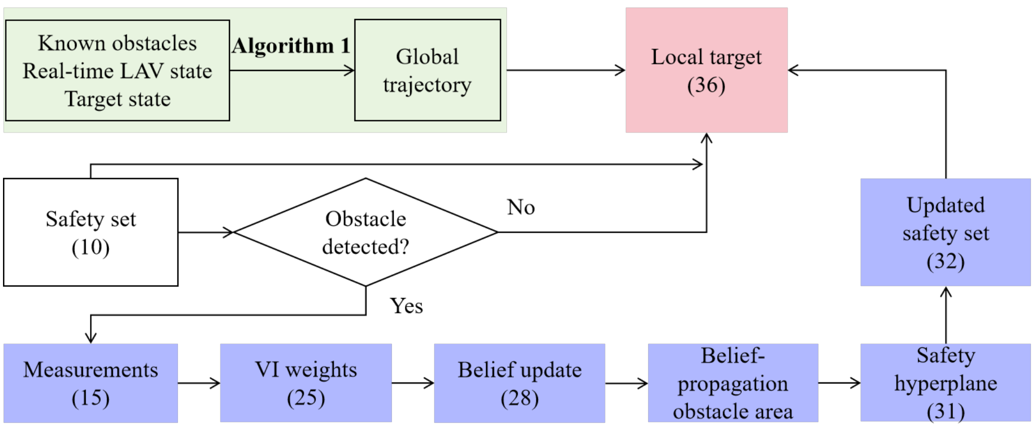

Figure 3. Workflow of the proposed safety control algorithm.

3.4. Theoretical Analysis of the Safety Control Algorithm

3.4.1. Computational Complexity and Scalability Analysis

The online computational load of the proposed algorithm stems from three main components: perception-driven belief propagation, hyperplane construction for each detected obstacle, and the event-triggered global trajectory optimization.

- For each obstacle within the initial safety set at time t, the belief state is updated using VI-based belief dynamics (Line 7 in Algorithm 2), followed by the construction of a separating hyperplane (Line 8). The belief update entails matrix operations with complexity $\mathcal{O}(d^3)$ (where d is the state dimension), whereas the hyperplane construction involves only 2D geometric projections and rotations with constant complexity $\mathcal{O}(1)$, incurring negligible overhead. Consequently, the total complexity for processing all $n_D(t)$ obstacles is $\mathcal{O}\big(n_D(t) \cdot d^3\big)$. This significantly outperforms the PF baseline complexity of $\mathcal{O}(n_D(t) \cdot N_\mathrm{P} \cdot d^2)$ by orders of magnitude, given that $N_\mathrm{P} \gg d$ (typically $N_\mathrm{P} \in [10^3, 10^4]$).
- The global optimization routine within Algorithm 1 is triggered only upon the detection of a new obstacle. Its worst-case complexity is $\mathcal{O}\big(S_{\max} \cdot (\log N_S + n_{\mathrm{kn}} \cdot N_L)\big)$, where $S_{\max}$ denotes the maximum sampling count, n_{kn} represents the number of known obstacles, N_S is the cardinality of the state node set $S_V(t)$, and N_L corresponds to the number of collision checkpoints sampled along the discretized trajectory (Line 8 in Algorithm 1).

The algorithm achieves efficient scalability in dense environments due to two key design features. First, the perception-related computational cost scales with $n_D(t)$ (the number of obstacles within the sensor's field of view at time t) rather than the total number of obstacles in the global environment. Notably, $n_D(t)$ is physically bounded by the sensor's field of view and the LAV's flight altitude. Second, the most computationally intensive component—global optimization—is event-triggered rather than periodic. Furthermore, the timeout parameter $S_{\max}$ imposes a hard upper bound on its execution time, thereby guaranteeing real-time performance even in complex scenarios.

3.4.2. Recursive Feasibility Analysis

The initial safety set in (10), denoted as $F_0(t)$, is intersected with a series of half-spaces to yield the updated safety set $F(t)$. This operation preserves convexity (see Appendix C). A critical observation is that the LAV's own position projection, $\boldsymbol{M}_2\boldsymbol{X}(t)$, which lies at the center of $F_0(t)$, cannot be excluded by any separating hyperplane. Consequently, $F(t)$ is guaranteed to be non-empty for all time t.

Given $\boldsymbol{M}_2\boldsymbol{X}(t) \in F(t)$ and the continuity of the global trajectory's x-y projection within $F(t)$, the existence of a feasible local target $\boldsymbol{X}_{\mathrm{lg}}(t)$ is guaranteed. Furthermore, the convexity of $F(t)$, combined with the inclusion of both the current state projection $\boldsymbol{M}_2\boldsymbol{X}(t)$ and the target projection $\boldsymbol{M}_2\boldsymbol{X}_{\mathrm{lg}}(t)$ within $F(t)$, ensures that the convex optimization problem formulated in (37) admits a feasible solution.

Solving (37) yields an optimal control sequence $\boldsymbol{u}_{t|k}$ ($k = 0, 1, \ldots, N-1$), where the first element $\boldsymbol{u}_{t|0}$ drives the system to the next state $\boldsymbol{X}(t+\tau)$. By the safety set constraint in (37), the projected next state satisfies $\boldsymbol{M}_2\boldsymbol{X}(t+\tau) \in F(t)$. Since $\boldsymbol{M}_2\boldsymbol{X}(t+\tau)$ serves as the center of $F_0(t+\tau)$ (the initial safety set at the next time step) and cannot be excluded by any separating hyperplane, it follows that $\boldsymbol{M}_2\boldsymbol{X}(t+\tau) \in F(t+\tau)$. This completes the inductive step, proving that a feasible state and corresponding control input exist at every subsequent time step, provided the optimization solver converges successfully at each iteration.

Remark on Numerical Robustness: In practical deployment, to mitigate the risk of solver non-convergence within the allotted time, the controller implements a Zero-Order Hold fallback strategy, reapplying the control input from the previous time step, $\boldsymbol{u}(t-\tau)$.

While this strategy does not guarantee collision avoidance under prolonged computational failure, it maintains system stability during transient delays, effectively granting the solver an additional cycle to recover feasibility.

3.4.3. Stability and Safety Analysis

We define the remaining path length to the global target $\boldsymbol{X}_\text{d}$ as a candidate Lyapunov-like function, $V_L(t) = L(t)$. Unlike standard asymptotic stability proofs where $\dot{V}_L(t) \leq 0$ must hold globally, our system operates in a dynamic environment where obstacle avoidance may temporarily increase the path length (i.e., $\dot{V}_L(t) > 0$). Therefore, we establish convergence based on piecewise monotonicity and bounded disturbances:

- Local Monotonicity: Between any two global replanning events (i.e., while the local target $\boldsymbol{X}_\text{lg}$ remains fixed), the local optimization in (37) guarantees that $\dot{V}_L(t) \leq 0$. Thus, the system strictly reduces the distance to the current local target.
- Bounded Increases: When a new unknown obstacle is detected, the global planner updates the trajectory, which may induce a finite, bounded increase in $V_L(t)$. Let this jump be denoted as $\Delta V_{L,\kappa} > 0$ at the κ-th replanning instant. This increase is inherently bounded by the sensor horizon and the geometric complexity of the environment.
- Finite-Time Convergence: Let t_κ denote the time of the κ-th replanning event. The total change in $V_L(t)$ over the mission duration $[0, t_f]$ can be expressed as

$$V_L(t_f) = V_L(0) + \sum_{\kappa} \Delta V_{L,\kappa} + \int_0^{t_f} \dot{V}_L(t)\,\mathrm{d}t, \tag{38}$$

Since the environment is bounded and the number of necessary replanning events to reach the goal is finite, the sum of positive jumps $\sum_\kappa \Delta V_{L,\kappa}$ is finite. Meanwhile, the integral term $\int_0^{t_f} \dot{V}_L(t)\,\mathrm{d}t$ represents the cumulative reduction in path length. Since this term is strictly negative during nominal execution and accumulates over time, it eventually dominates the sum of finite positive jumps as the LAV approaches the target. Consequently, $V_L(t)$ is guaranteed to reach the predefined threshold ϵ in finite time. Consistent with Algorithm 1, once $V_L(t) \leq \epsilon$, the LAV is considered to have reached the target $\boldsymbol{X}_\text{d}$, and the task is immediately terminated. This concludes the proof of finite-time convergence to the target.

The adaptive safety set update rule in (32) provides a probabilistic safety guarantee. For each obstacle within the initial safety set at time t, a buffered separating hyperplane defined by $\bar{\boldsymbol{A}}_j[x(t), y(t)]^\top = \bar{b}_j - b_j^\delta$ is constructed, where the buffer b_j^δ is derived from the collision probability threshold δ_o to explicitly account for relative state uncertainty. Enforcing the constraint $\bar{\boldsymbol{A}}_j[x(t), y(t)]^\top \geq \bar{b}_j - b_j^\delta$ guarantees that the collision probability between the LAV and the obstacle is upper-bounded by δ_o. Since $\boldsymbol{M}_2\boldsymbol{X}(t) \in F(t)$ and $F(t)$ incorporates all such half-space constraints for detected obstacles, the collision probability for all detected obstacles remains below δ_o for all t. This establishes a formal guarantee of probabilistic safety.

4. Simulation Results

To validate the proposed algorithm, a simulation scenario is established within a bounded airspace containing eight distinct obstacles. These obstacles are classified as either known or unknown using a binary indicator, where 1 denotes a known obstacle and 0 indicates an unknown one. The detailed geometric and positional parameters of the obstacles are listed in Table 1, while the algorithm parameters and simulation settings are summarized in Table 2.

Table 1. Obstacle parameters.

	Initial Position (m)	Input (m/s²)	$\varepsilon_{\mathrm{o}}^{j}$ (m)	Flag
Obstacle 1	[3500, 2000]	[0, 0, 0]	[400, 400]	1
Obstacle 2	[2500, 1000]	[0, 0, 0]	[400, 300]	1
Obstacle 3	[2500, 2500]	[−0.2, −0.2, 0]	[300, 400]	0
Obstacle 4	[1000, 500]	[0.2, 0.2, 0]	[300, 300]	0
Obstacle 5	[1500, 1500]	[0, 0, 0]	150	1
Obstacle 6	[2000, 2000]	[0, 0, 0]	200	1
Obstacle 7	[2500, 2000]	[0.2, −0.2, 0]	200	0
Obstacle 8	[3500, 1000]	[−0.2, 0.2, 0]	150	0

Table 2. Experimental setup parameters.

Symbol	Value	Symbol	Value	CasADi Parameters	Value
α	70°	$\boldsymbol{X}_0$	[500 m, 1500 m, 200 m, 30 m/s, 0, 0]	max_iter	2000
β	70°	$\boldsymbol{X}_{\mathrm{d}}$	[4500 m, 1500 m, 200 m, 30 m/s, 0, 0]	print_level	0
δ_{o}	0.1	$\boldsymbol{Q}$	diag(5, 5, 5, 0, 0, 0)	acceptable_tol	1×10^{-8}
M	3	$\boldsymbol{R}$	diag(1, 1, 1)	acceptable_obj_change_tol	1×10^{-6}
$S_{\max}$	500	$\boldsymbol{\Sigma}_0$	diag(25 m, 25 m, 0, 4 m/s, 4 m/s, 0)	print_time	0
ϵ	50 m	$\boldsymbol{Q}_{\mathrm{o}}(t)$	diag(0.04, 0.04, 0) $\mathrm{m/s^2}$	nlpsol	'ipopt'
τ	0.1 s	$\alpha_{m,t}$	0.6, 0.3, 0.1		
n_{kn}	4	$\boldsymbol{\mu}_{m,t}$	[0.5 m; 0; 0], [0; 0.1 rad; 0], [−0.5 m; −0.1 rad; 0]		
N	10	$\boldsymbol{R}_{m,t}$	diag(1 m, 0.01 rad, 0), diga(5 m, 0.05 rad, 0), diag(10 m, 0.1 rad, 0)		
$\gamma(t)$	$[-\pi/6, \pi/6]$	χ	[0, 5000] m × [0, 3000] m × [100, 300] m × [20, 30] m/s × [−30, 30] m/s × [−30, 30] m/s		
$\psi(t)$	$[-\pi/4, \pi/4]$	$\mathcal{U}$	$[-10, 10]\ \mathrm{m/s^2} \times [-10, 10]\ \mathrm{m/s^2} \times [-10, 10]\ \mathrm{m/s^2}$		

4.1. Results and Analysis

The LAV is equipped with dual onboard sensors covering both frontal and lateral sectors, each with a 70° field of view, to enable real-time obstacle detection. As shown in Figure 4, a binary detection indicator reflects the dynamic interaction between the LAV and obstacles: a value of 1 signifies that an obstacle is within the detection range, while 0 indicates its absence. Obstacle 5 is the first to enter the detection zone at $t = 15.4$ s, followed sequentially by Obstacles 6, 3, 8, and 1. Notably, at no point do more than two obstacles appear simultaneously within the detection area.

Using Obstacle 5 as a representative case, the proposed algorithm leverages real-time measurements to estimate the obstacle's position immediately upon detection; the corresponding position estimation error is depicted in Figure 5. Subsequently, the algorithm identifies the point on the belief-based obstacle region closest to the LAV and constructs a separating hyperplane (represented as a line in the x-y plane), as illustrated in Figure 6. The safety set is then updated by intersecting the initial safety set with the half-space defined by a normal vector pointing from this nearest point to the LAV's projected position. This process facilitates real-time obstacle avoidance and the dynamic adaptation of the safety set.

Local targets are defined as the intersection points between the current safety set (initial or updated) and the global trajectory. Transitions between these targets are marked along the LAV's flight path in Figure 7. The deviations observed between the local targets and the actual trajectory are attributed to the linear predictive model used in the local optimization formulation. Throughout the mission, the LAV executes ten global optimizations, each associated with a distinct local target. Two significant deviations occur in the latter half of the mission, directly resulting from avoidance maneuvers performed when the LAV simultaneously encounters the known Obstacle 1 and the unknown, moving Obstacle 8.

By integrating obstacle detection, state estimation, safety set updates, and local target generation, the LAV maintains reliable operation in obstacle-free regions throughout the mission. This performance is quantitatively verified by the minimum separation distance between the LAV and all obstacles, which remains strictly positive for the entire duration (Figure 8). Furthermore, the robustness of the proposed approach is validated through 1000 Monte Carlo simulation trials. In 954 of these trials, the minimum separation distance remained consistently positive (Figure 9), yielding an empirical success rate of 95.4%. This result satisfies the specified probabilistic safety guarantee, as $95.4\% > (1 - \delta_\mathrm{o}) \times 100\%$.

The remaining 4.6% of failures are attributed to the underestimation of state covariance when approximating the GMM likelihood as a single Gaussian. This approximation introduces an optimistic bias, resulting in a safety buffer (b_j^δ) that is insufficient to cover the heavy tails of the true distribution, thereby leading to unanticipated constraint violations under extreme noise realizations. Across the 1000 Monte Carlo simulations, the average discrepancy between the proposed buffer b_j^δ and a conservative Chebyshev-based buffer (b_j^{Cheb}) was found to be 10.6 m, quantifying the magnitude of this optimistic bias.

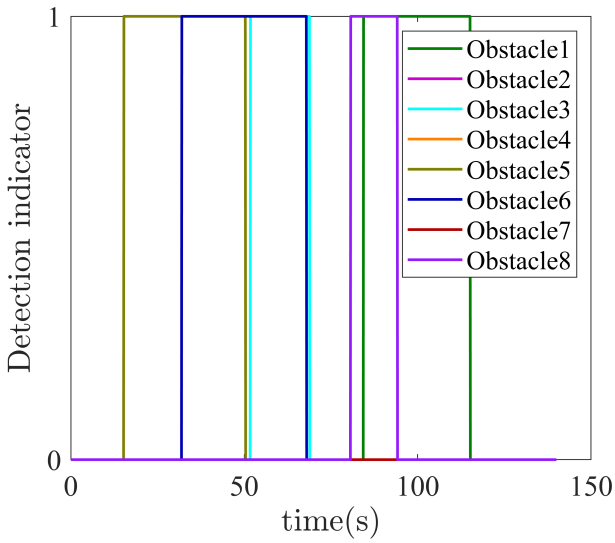

Figure 4. Timeline of obstacle detection events (binary indicator: 1 for detected, 0 for undetected).

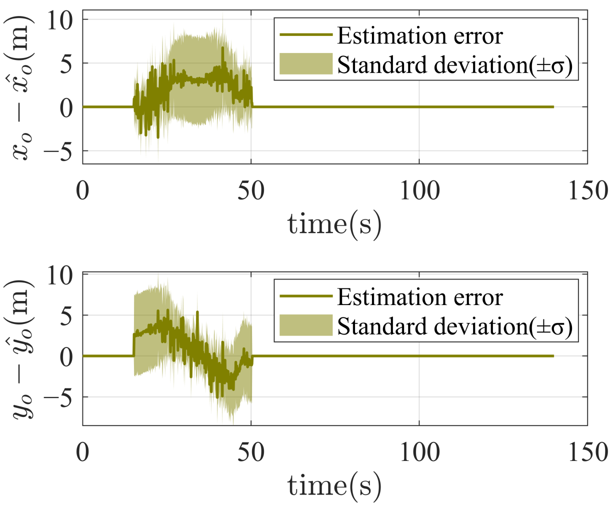

Figure 5. Position estimation error for Obstacle 5 (the first detected obstacle).

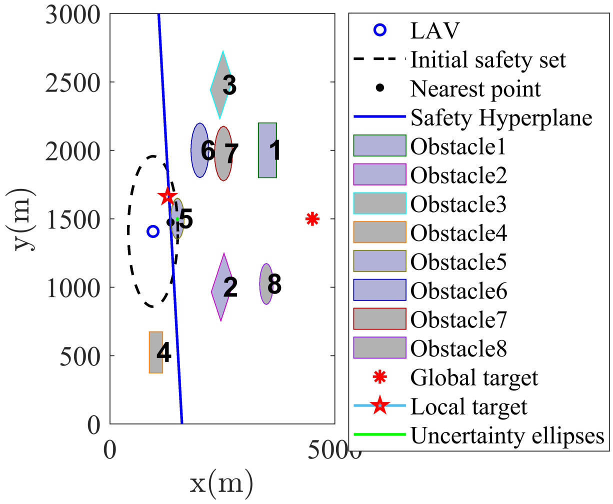

Figure 6. Adaptive update of the safety set triggered by Obstacle 5 at t =15.4 s.

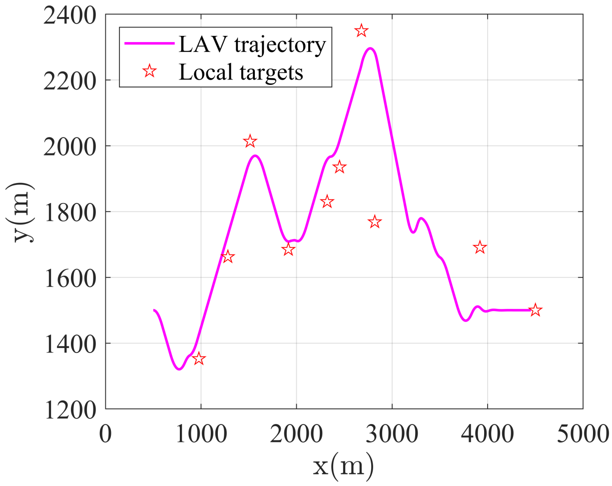

Figure 7. LAV flight trajectory showing sequentially generated local targets.

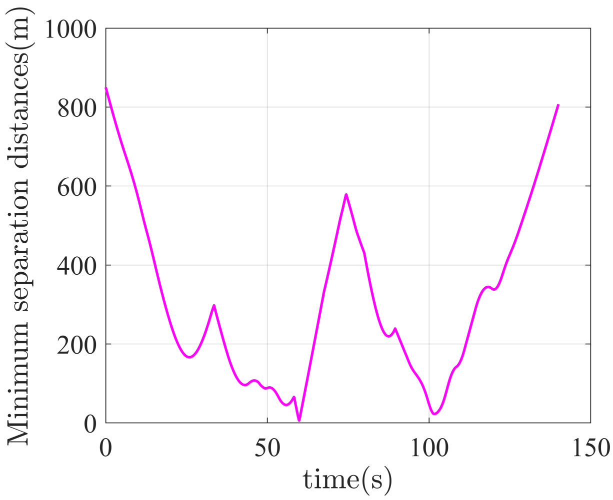

Figure 8. Time history of the minimum separation distance in a representative simulation trial.

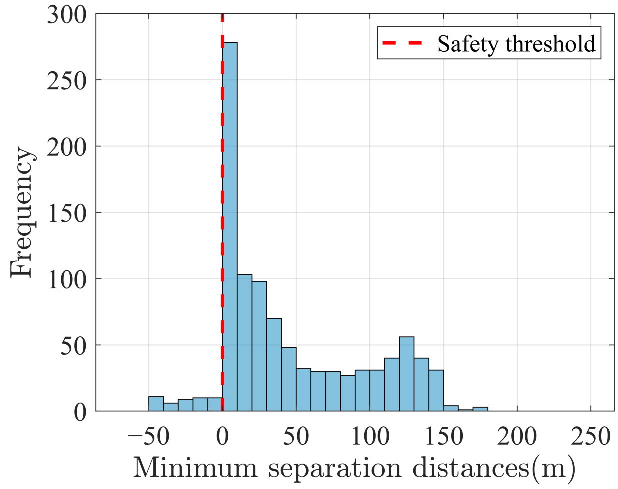

Figure 9. Statistical distribution of the minimum separation distance across 1000 Monte Carlo trials.

4.2. Comparative Analysis

4.2.1. Estimation Performance Comparison

The performance of the presented VI-based estimation method is compared with that of the multi-model EKF (MMEKF) to validate the fidelity of the Gaussian approximation. In the MMEKF framework, filters corresponding to each Gaussian component are updated in parallel, and their posterior distributions are fused into a single Gaussian distribution via weighting

$$\alpha_{m,t} = \frac{1}{\sqrt{\det(2\pi S_{m,t})}} \exp\left(-\frac{1}{2} z_{m,t}^{\top} S_{m,t}^{-1} z_{m,t}\right),$$
$$S_{m,t} = H_t \Sigma_t^{-1} H_t^{\top} + R_{m,t}, \quad z_{m,t} = z_t - \mu_{m,t} - \bar{h}\left(\hat{X}_t^{-}\right).$$

The root-mean-square error (RMSE) and average negative log-likelihood (NLL) were calculated for all detected obstacles, with the results summarized in Table 3. The comparison reveals negligible differences in RMSE (<5 cm) and NLL (<0.2) between the two methods. These results confirm that the proposed VI approach successfully captures the dominant characteristics of the measurement distribution without significant fidelity loss.

Unlike the MMEKF, which yields a multi-modal posterior (Gaussian mixture), the proposed VI method explicitly approximates the non-Gaussian likelihood as a single Gaussian component, thereby ensuring a unimodal posterior. This structural simplification is critical: it eliminates the need for complex multi-modal constraint handling in the downstream controller. Consequently, the slight increase in computational cost (0.12 ms) represents a negligible trade-off for guaranteeing the mathematical tractability and real-time feasibility of the overall navigation framework.

Table 3. Performance comparison of the two estimation methods.

	RMSE (m)		Average NLL		Average Calculation Time (ms)	
	MMEKF	VI	MMEKF	VI	MMEKF	VI
Obstacle 1	3.18	3.14	4.65	4.76		
Obstacle 3	3.36	3.32	4.68	4.77		
Obstacle 5	3.78	3.71	4.82	4.89	0.30	0.42
Obstacle 6	3.45	3.45	4.66	4.82		
Obstacle 8	3.66	3.47	4.82	4.81		

4.2.2. Control Performance Comparison

To evaluate the performance of the proposed algorithm (Method 1), a comparative study was conducted against the robust tube MPC approach from [24] (Method 2) and the uncertainty-aware Voronoi-based method from [27] (Method 3). The extended LAV flight duration (approximately two minutes), combined with a short step size (0.1 s), imposes an impractically long prediction horizon on Method 2 if it is to satisfy its terminal constraints directly. To adapt Method 2 to the present scenario, it was modified by incorporating the local target generation strategy from Method 1 and configured with distinct disturbance bounds. The detailed extensions and parameter settings for the Method 2 variants are provided in Table 4. Method 3 identifies a local goal within a buffered uncertainty-aware Voronoi cell—functionally analogous to the safety set—by solving a quadratic programming problem subject to linear inequality constraints. To satisfy the solver requirements of Method 3 and, crucially, to ensure a fair comparison under identical initial conditions for all algorithms, the initial safety set for all methods was defined as the maximum inscribed rectangle within the elliptical detection area. This adaptation converts the nonlinear elliptical constraints into linear box constraints, enabling the feasible implementation of Method 3. All such adaptations and parameter tuning steps were implemented to eliminate biases stemming from platform and scenario disparities, thereby ensuring that the performance comparison accurately reflects the efficacy of each algorithm's core logic.

Table 4. Extended method 2: implementation details and parameter settings.

	Prediction Horizon N	Disturbance Bounds x (m)	Disturbance Bounds y (m)	Use of Local Target Generation Strategy?
Method 2-1	1500	$[-0.2, 0.2]$	$[-0.2, 0.2]$	No
Method 2-2	10	$[-0.2, 0.2]$	$[-0.2, 0.2]$	Yes
Method 2-3	10	$[-5.0, 5.0]$	$[-5.0, 5.0]$	Yes

In addition to the above safety-set-based approaches, this section also includes comparisons with two baseline methods: deterministic MPC and stochastic MPC. For both baselines, the prediction horizon is set to $N = 10$. The key distinction between these baselines and the safety-set-based methods lies in how obstacle avoidance is enforced: while the latter constrain the LAV to remain within a predefined safe region, the former treat collision avoidance as hard constraints directly imposed on the trajectory. Specifically, the deterministic MPC enforces the constraint $f\left(x(t), y(t), \hat{O}_{\mathrm{o}}^{j}(t)\right) \geq 0$, whereas the stochastic MPC requires $\mathbb{P}\left\{f\left(x(t), y(t), \hat{O}_{\mathrm{o}}^{j}(t)\right) \geq 0\right\} \geq 1 - \delta_{\mathrm{o}}$.

Obstacle avoidance performance across all methods in the specified scenario is analyzed and compared based on LAV flight trajectories (Figure 10) and minimum separation distance curves (Figure 11). For Method 2-1 (the original Method 2), the target state is fixed ($\boldsymbol{X}_{\mathrm{d}}$), and no feasible solution can be found when the target state lies outside the collision avoidance half-space. To address this limitation, the local target generation strategy from Method 1 was integrated into Method 2. The results show that Method 2-2, configured with a small disturbance bound designed to match the process noise level, successfully evades all obstacles. In contrast, Method 2-3—with a larger disturbance bound designed according to the initial error standard deviation—fails due to excessive conservatism. This conservatism contracts the admissible input space (specifically, to $\mathcal{U} = [-5.98, 5.98] \times [-5.98, 5.98] \times [-10, 10]\ \mathrm{m/s^2}$), resulting in insufficient control authority for obstacle avoidance maneuvers. For Method 3, obstacle avoidance is successfully achieved in the early planning phase; however, the method fails in the later stage when the LAV is simultaneously confronted with Obstacles 1, 7, and 8. The root cause is the myopic nature of its local goal planning, which lacks sufficient foresight for global coordination in complex obstacle scenarios and thus traps the LAV in a local optimum. In comparison to the safety-set-based methods, the baseline methods exhibit a delayed response to the known Obstacle 5. This latency arises because their avoidance constraints are triggered only when the predicted future position enters the obstacle area, leading to a brief intrusion of the LAV into the threat zone of Obstacle 5 (i.e., the minimum separation distance drops below zero).

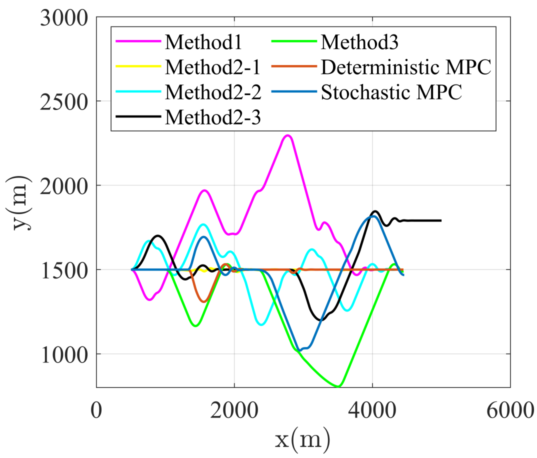

Figure 10. LAV flight trajectories generated by all control methods.

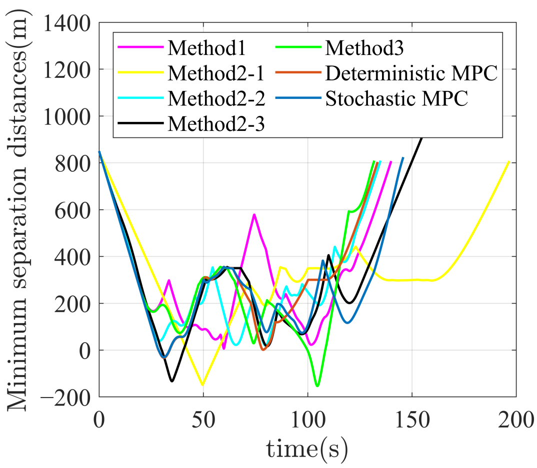

Figure 11. Time history of the minimum separation distance for all control methods.

Finally, the proposed algorithm is compared with the other four methods across additional performance metrics, as summarized in Table 5. Real-time performance is evaluated by measuring the average computation time per complete planning cycle, as defined in Algorithm 2, under a consistent computational environment: Python 3.8 running on a machine equipped with a 12-core Intel Core i7-12700H processor. As shown in Figure 12, both the proposed Method 1 and the extended Method 2-2 occasionally trigger global trajectory optimization, resulting in higher computation times compared to Method 3, which relies solely on local planning within a safety set. While the inclusion of global optimization incurs higher control effort and computation time, it proves decisive in achieving a high success rate. Notably, the average computation time of Method 1 is 0.0256 s (approximately 39 Hz), demonstrating robust onboard feasibility. This performance is well within the capability of mid-tier airborne computers commonly used in unmanned aerial vehicles, such as the NVIDIA Jetson TX2/NX series and the Raspberry Pi 4. Consequently, the proposed algorithm can be deployed on standard commercial hardware without requiring specialized hardware or ground-station offloading.

Table 5. Comparative performance metrics of the five control methods: computation time, control effort, path length, and success rate.

	Average Computation Time (s)	Control Effort (m^2/s^4)	Path Length (m)	Success Rate in 1000 Trials
Method 1	0.0256	65,554.8	5000.4	95.4%
Method 2-2	0.0507	79,806.7	4879.2	98.4%
Method 3	0.0192	36,287.1	4762.1	0
Deterministic MPC	0.0200	26,657.0	4134.0	0
Stochastic MPC	0.0330	38,783.4	4769.7	0

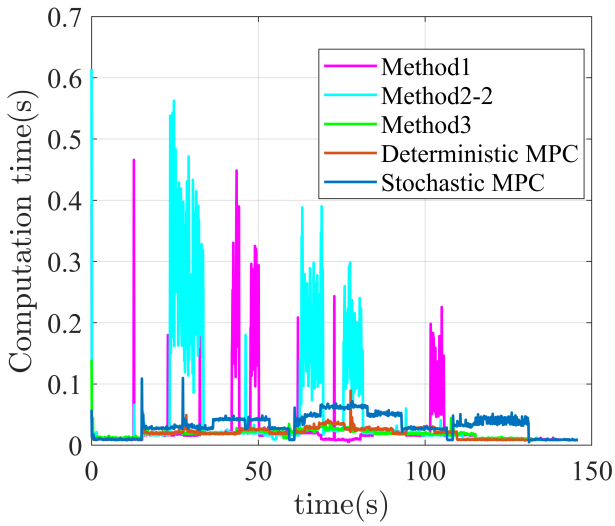

Figure 12. Computation time per planning cycle for the five methods over the mission duration.

4.3. Dense Obstacle Scenario

This section validates the scalability of the proposed algorithm in a dense environment containing 30 obstacles, a significant increase from the sparse scenarios (8 obstacles) discussed previously. The obstacles are randomly generated within the spatial region $[1500, 4000] \times [0, 3000]$ m. To simulate dynamic uncertainty, the acceleration of each obstacle is randomized at every control cycle within $[-2, 2] \times [-2, 2]$ m/s^2, while their sizes are uniformly sampled from $[100, 300]$ m. Furthermore, to test the adaptive perception capability, each obstacle is stochastically assigned a label of 0 (unknown) or 1 (known).

To explicitly induce a challenging multimodal posterior distribution with sharper peaks than those in Table 2, the GMM parameters are configured with reduced covariance values. The means are set to:

$$\boldsymbol{\mu}_{1,t} = \begin{bmatrix} 2.0\ \text{m} \\ 0.3\ \text{rad} \\ 0 \end{bmatrix}, \quad \boldsymbol{\mu}_{2,t} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \boldsymbol{\mu}_{3,t} = \begin{bmatrix} -2.0\ \text{m} \\ -0.3\ \text{rad} \\ 0 \end{bmatrix},$$

and the covariance matrices are defined as diagonal structures:

$$\boldsymbol{R}_{1,t} = \text{diag}(0.08\text{m}, 0.02\text{rad}, 0), \boldsymbol{R}_{2,t} = \text{diag}(0.10\text{m}, 0.03\text{rad}, 0), \boldsymbol{R}_{3,t} = \text{diag}(0.12\text{m}, 0.04\text{rad}, 0).$$

This configuration creates a true measurement posterior with clearly separated, high-confidence modes, as illustrated in Figure 13.

In this highly congested scenario, the LAV simultaneously detects and tracks up to 7 obstacles within its field of view, as shown in Figure 14. The minimum separation distance between the LAV and the nearest obstacle (Figure 15) confirms that the algorithm successfully avoids all collisions, maintaining rigorous safety constraints throughout the trajectory.

These results offer an interesting perspective on the limitations and potential of the single-Gaussian approximation. While this approximation inherently smooths over distinctly separated multimodal beliefs, the success observed in this specific configuration suggests that *mission failure is not an inevitable consequence of multimodality*. It appears that when the Gaussian smoothing primarily affects the low-probability valleys between modes—leaving the high-probability mass defining the core obstacle locations largely intact—the approximated posterior can remain sufficiently representative of the true risk. This observation invites further discussion: rather than strictly requiring complex multimodal representations, ensuring that adaptive safety buffers encompass the smoothed yet conservative belief might be a computationally efficient alternative for handling severe multimodality in dense environments.

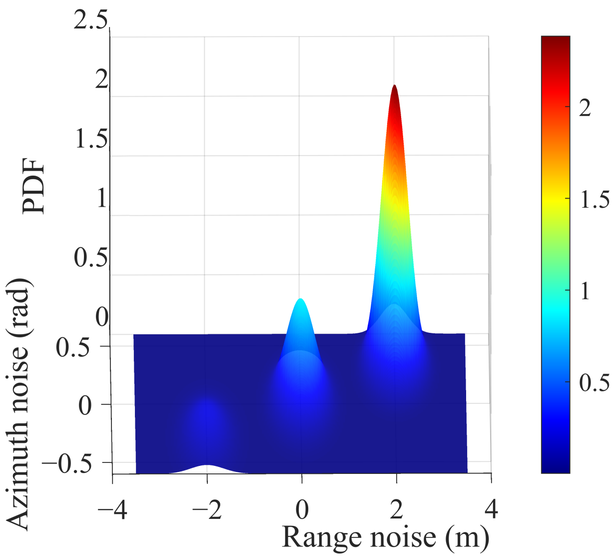

Figure 13. Spatial distribution of the true multimodal measurement posterior in the dense scenario.

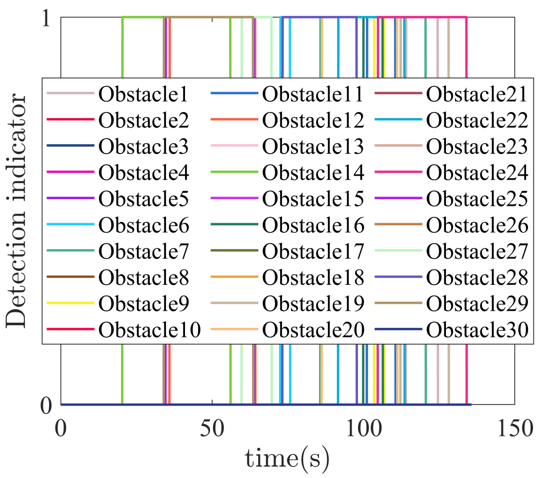

Figure 14. Temporal evolution of the number of detected obstacles in the dense scenario.

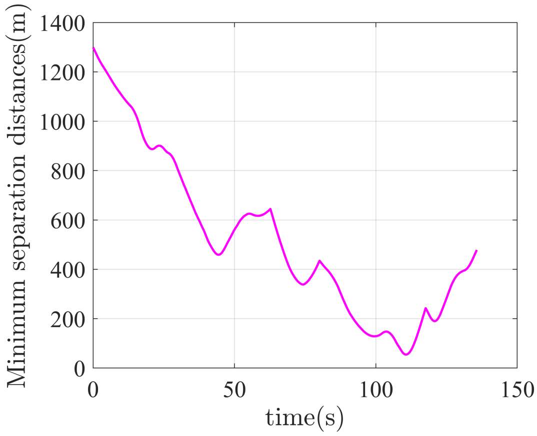

Figure 15. Temporal profile of the minimum separation distance ensuring collision avoidance.

5. Discussion

This section summarizes the study's core findings, contextualizes them within existing research, and outlines limitations and future directions.

This study proposed a safety control algorithm for LAVs that integrates an adaptive safety set with belief dynamics. Extensive Monte Carlo simulations (1000 trials) confirmed the algorithm's capability to enable reliable real-time obstacle avoidance in mixed known/unknown obstacle environments under relative state uncertainty. Specifically, the method achieved a 95.4% safety success rate with a collision probability threshold of 0.1. Furthermore, qualitative validation in a dense scenario with 30 obstacles demonstrated the algorithm's scalability, maintaining collision-free trajectories even when tracking multiple objects simultaneously under severe multimodal measurement noise.

Comparative evaluations against state-of-the-art methods—including MMEKF, robust MPC, and uncertainty-aware Voronoi-based approaches—as well as baseline strategies, illustrate the performance of the proposed algorithm. In terms of state estimation, while achieving RMSE and NLL metrics comparable to those of MMEKF, the proposed method uniquely provides a unimodal Gaussian posterior. This characteristic is pivotal for facilitating rapid safety control via convex safety sets. Regarding obstacle avoidance, the algorithm effectively leverages global information and uncertainty distributions to excel in complex scenarios. It overcomes the limitations of robust MPC (specifically, computationally expensive prediction horizons and heavy reliance on disturbance bounds) and addresses the myopic planning issues inherent in Voronoi-based methods within multi-obstacle environments.

This study offers significant theoretical and practical contributions. Theoretically, it provides a modular design reference for autonomous navigation systems by integrating VI-based Gaussian approximation with dynamic safety set updates. Practically, its lightweight nature (operating at an update frequency of 39 Hz) and compatibility with generic relative-state sensors (requiring only relative distance and angle inputs) facilitate deployment on resource-constrained LAV platforms.

However, critical limitations remain: (i) The VI-based estimation approximates the GMM measurement likelihood as a single Gaussian. While our dense scenario results suggest that this smoothing does not necessarily compromise safety if high-probability modes are preserved, this approach inherently struggles with strongly multimodal or heavy-tailed distributions where critical spatial information might be lost, potentially degrading estimation performance under significant measurement anomalies. (ii) The algorithm has been validated only for static and stochastically moving obstacles; its adaptability and performance have not yet been tested against obstacles exhibiting active maneuvers with adversarial behaviors.

Future research will address these identified limitations. First, we plan to incorporate an event-triggered mechanism into the estimator module (e.g., switching to moment matching or advanced nonlinear estimators when strong multimodality is detected) to enhance robustness against measurement anomalies while retaining the computational efficiency of the Gaussian approximation in nominal conditions. Second, we will extend the algorithm to scenarios involving obstacles capable of adaptive active maneuvers. Finally, we aim to develop adaptive mechanisms, such as dynamically adjusting the probabilistic safety buffer based on the local density of detected obstacles, to further optimize computational efficiency and avoidance reliability.

Author Contributions: Conceptualization, Y.W. and W.Z.; methodology, M.Z.; software, M.Z.; validation, M.Z. and Y.W.; formal analysis, Y.W.; investigation, M.Z.; resources, W.Z.; data curation, M.Z.; writing—original draft preparation, M.Z.; writing—review and editing, Y.W.; visualization, M.Z.; supervision, W.Z.; project administration, Y.W.; funding acquisition, W.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This work was funded by the National Natural Science Foundation of China grant number 62373366, and the Provincial Science Fund for Distinguished Young Scholars of Hunan grant number 2024JJ2064.

Data Availability Statement: The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest: The authors declare no conflict of interest.

Appendix A. Linearization of the Nonlinear Mapping Function

The onboard sensors of the LAV provide relative measurements of range, azimuth, and elevation to obstacles. The corresponding nonlinear measurement function is given by

$$\bar{h}(\bar{\boldsymbol{X}}_t) = \begin{bmatrix} \sqrt{(x_o^j(t) - x(t))^2 + (y_o^j(t) - y(t))^2 + (-h(t))^2} \\ \arctan\left(\dfrac{y_o^j(t) - y(t)}{x_o^j(t) - x(t)}\right) \\ \arctan\left(\dfrac{-h(t)}{\sqrt{(x_o^j(t) - x(t))^2 + (y_o^j(t) - y(t))^2}}\right) \end{bmatrix}. \tag{A1}$$

Performing a first-order Taylor expansion of $\bar{h}(\bar{\boldsymbol{X}}_t)$ about the prior state estimate $\hat{\bar{\boldsymbol{X}}}_t^-$ yields

$$\bar{h}(\bar{\boldsymbol{X}}_t) \approx \bar{h}(\hat{\bar{\boldsymbol{X}}}_t^-) + \boldsymbol{H}_t(\bar{\boldsymbol{X}}_t - \hat{\bar{\boldsymbol{X}}}_t^-) \tag{A2}$$

where $\boldsymbol{H}_t = \dfrac{\partial \bar{h}}{\partial \bar{\boldsymbol{X}}}\Big|_{\bar{\boldsymbol{X}}=\hat{\bar{\boldsymbol{X}}}_t^-}$ is the linearized measurement Jacobian matrix evaluated at the operating point $\hat{\bar{\boldsymbol{X}}}_t^-$. The explicit form of $\boldsymbol{H}_t$ is

$$\boldsymbol{H}_t = \begin{bmatrix} h_{11} & h_{12} & h_{13} & 0 & 0 & 0 \\ h_{21} & h_{22} & 0 & 0 & 0 & 0 \\ h_{31} & h_{32} & h_{33} & 0 & 0 & 0 \end{bmatrix}, \tag{A3}$$

with the elements defined as follows

$$
\begin{aligned}
h_{11} &= \frac{\hat{x}^-}{\sqrt{(\hat{x}^-)^2 + (\hat{y}^-)^2 + (\hat{h}^-)^2}}, & h_{12} &= \frac{\hat{y}^-}{\sqrt{(\hat{x}^-)^2 + (\hat{y}^-)^2 + (\hat{h}^-)^2}}, & h_{13} &= \frac{\hat{h}^-}{\sqrt{(\hat{x}^-)^2 + (\hat{y}^-)^2 + (\hat{h}^-)^2}}, \\
h_{21} &= \frac{-\hat{y}^-}{(\hat{x}^-)^2 + (\hat{y}^-)^2}, & h_{22} &= \frac{\hat{x}^-}{(\hat{x}^-)^2 + (\hat{y}^-)^2}, \\
h_{31} &= \frac{-\hat{x}^-\hat{h}^-\left[(\hat{x}^-)^2 + (\hat{y}^-)^2\right]^{-1/2}}{(\hat{x}^-)^2 + (\hat{y}^-)^2 + (\hat{h}^-)^2}, & h_{32} &= \frac{-\hat{y}^-\hat{h}^-\left[(\hat{x}^-)^2 + (\hat{y}^-)^2\right]^{-1/2}}{(\hat{x}^-)^2 + (\hat{y}^-)^2 + (\hat{h}^-)^2}, & h_{33} &= \frac{\left[(\hat{x}^-)^2 + (\hat{y}^-)^2\right]^{1/2}}{(\hat{x}^-)^2 + (\hat{y}^-)^2 + (\hat{h}^-)^2}
\end{aligned}
\tag{A4}
$$

where $\hat{\bar{\boldsymbol{X}}}_t^- = [\hat{x}^-, \hat{y}^-, \hat{h}^-, \hat{\dot{x}}^-, \hat{\dot{y}}^-, \hat{\dot{h}}^-]^\top$ denotes the prior estimate of the relative state between the LAV and the obstacle.

Appendix B. Closed-Form Solution of the Variational Weight

To address the constrained optimization problem in (24), we devise the Lagrangian formulation

$$
J = L(q) + \lambda \left(1 - \sum_{m=1}^{M} \gamma_{m,t}\right) \tag{A5}
$$

where λ denotes the Lagrange multiplier. Substituting the expanded ELBO from (23) into the above equation, the gradient of the Lagrangian J with respect to the variable $\gamma_{m,t}$ is

$$
\frac{\partial J}{\partial \gamma_{m,t}} = \log\left(\frac{\alpha_{m,t} \cdot \mathcal{N}(\boldsymbol{z}_t; \bar{h}(\bar{\boldsymbol{X}}_t) + \boldsymbol{\mu}_{m,t}, \boldsymbol{R}_{m,t})}{\gamma_{m,t}}\right) - 1 - \lambda. \tag{A6}
$$

By equating the gradient to zero to find the stationary point

$$
\gamma_{m,t} = \alpha_{m,t} \cdot \mathcal{N}(\boldsymbol{z}_t; \bar{h}(\bar{\boldsymbol{X}}_t) + \boldsymbol{\mu}_{m,t}, \boldsymbol{R}_{m,t}) \cdot \exp(-1 - \lambda). \tag{A7}
$$

The Lagrange multiplier term $\exp(-1-\lambda)$ acts as a normalization constant. It is solved by applying the constraint $\sum_{m=1}^{M} \gamma_{m,t} = 1$

$$
\exp(-1-\lambda) = \frac{1}{\sum_{m=1}^{M} \alpha_{m,t} \cdot \mathcal{N}(\boldsymbol{z}_t; \bar{h}(\bar{\boldsymbol{X}}_t) + \boldsymbol{\mu}_{m,t}, \boldsymbol{R}_{m,t})}. \tag{A8}
$$

Finally, substituting $\exp(-1-\lambda)$ back into the expression for $\gamma_{m,t}$ yields the closed-form solution presented in (25)

$$
\gamma_{m,t} = \frac{\alpha_{m,t} \cdot \mathcal{N}(\boldsymbol{z}_t; \bar{h}(\bar{\boldsymbol{X}}_t) + \boldsymbol{\mu}_{m,t}, \boldsymbol{R}_{m,t})}{\sum_{m=1}^{M} \alpha_{m,t} \cdot \mathcal{N}(\boldsymbol{z}_t; \bar{h}(\bar{\boldsymbol{X}}_t) + \boldsymbol{\mu}_{m,t}, \boldsymbol{R}_{m,t})}. \tag{A9}
$$

This derivation confirms that the optimal weights are simply the normalized likelihoods of the measurement under each Gaussian component.

Appendix C. Proof of Convexity for the Adaptive Safety Set

Given the defining function of the initial safety set

$$
f_D(x_D(t), y_D(t)) = \frac{(x_D(t) - x(t))^2}{L_1^2} + \frac{(y_D(t) - y(t))^2}{L_2^2}, \tag{A10}
$$

its Hessian matrix is given by

$$\nabla^2 f_D = \begin{bmatrix} 2/L_1^2 & 0 \\ 0 & 2/L_2^2 \end{bmatrix} \succ 0, \tag{A11}$$

which is positive definite. Therefore, f_D is a strictly convex function.

Consider any two points $\boldsymbol{p}_1, \boldsymbol{p}_2 \in F(t)$, where $\boldsymbol{p}_1 = [x_1(t), y_1(t)]^\top$ and $\boldsymbol{p}_2 = [x_2(t), y_2(t)]^\top$. By the definition of the set, it holds that $f_D(\boldsymbol{p}_1) \leq 1$ and $f_D(\boldsymbol{p}_2) \leq 1$. By the convexity of f_D, for any $\theta \in [0, 1]$, we have

$$f_D(\theta \boldsymbol{p}_1 + (1-\theta)\boldsymbol{p}_2) \leq \theta f_D(\boldsymbol{p}_1) + (1-\theta) f_D(\boldsymbol{p}_2) \leq \theta \cdot 1 + (1-\theta) \cdot 1 = 1. \tag{A12}$$

Hence, $\theta \boldsymbol{p}_1 + (1-\theta)\boldsymbol{p}_2 \in F(t)$, proving that the set $F(t)$ (an ellipse) is convex.

The updated safety set in (32) introduces additional constraints of the form

$$\bar{\boldsymbol{A}}_j \begin{bmatrix} x_D(t) \\ y_D(t) \end{bmatrix} \geq \bar{b}_j - b_j^\delta, \quad j = 1, \ldots, n_D(t). \tag{A13}$$

Each such inequality describes a closed half-space

$$H_j^{\mathrm{half}}(t) := \left\{ \boldsymbol{p} \in \mathbb{R}^2 \mid \bar{\boldsymbol{A}}_j \boldsymbol{p} \geq c_j \right\}, \quad c_j = \bar{b}_j - b_j^\delta. \tag{A14}$$

For any two points $\boldsymbol{p}_1, \boldsymbol{p}_2 \in H_j^{\mathrm{half}}(t)$ and any $\theta \in [0, 1]$, it holds that

$$\bar{\boldsymbol{A}}_j(\theta \boldsymbol{p}_1 + (1-\theta)\boldsymbol{p}_2) = \theta \bar{\boldsymbol{A}}_j \boldsymbol{p}_1 + (1-\theta)\bar{\boldsymbol{A}}_j \boldsymbol{p}_2 \geq \theta \cdot c_j + (1-\theta) \cdot c_j = c_j. \tag{A15}$$

Thus, $\theta \boldsymbol{p}_1 + (1-\theta)\boldsymbol{p}_2 \in H_j^{\mathrm{half}}(t)$, confirming that each $H_j^{\mathrm{half}}(t)$ is convex.

Since $F(t)$ (an ellipse) and each $H_j^{\mathrm{half}}(t)$ (a half-space) are convex sets, and the intersection of any collection of convex sets remains convex (a standard result in convex analysis), it follows that the updated safety set defined in (32) is convex.

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